NETWORKING

DNS Resolution Visual Guide

From example.com to 93.184.216.34
in milliseconds — the phonebook of the internet.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Problem: Humans Hate Numbers

You type a name; the network needs an IP address.

example.com

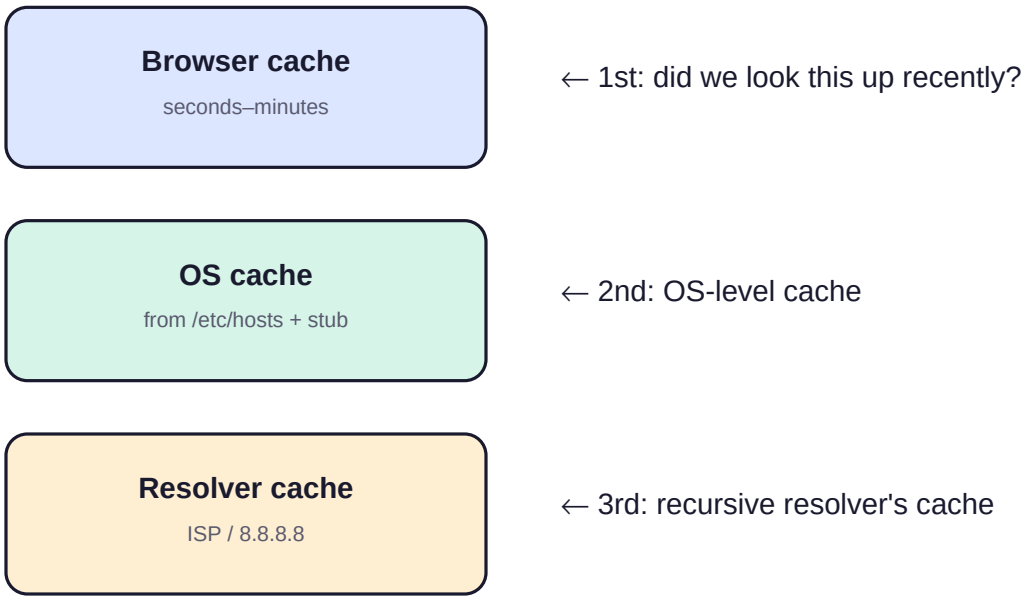
`93.184.216.34`

DNS is a hierarchical, distributed database mapping names → IPs (and more). ~trillions of queries per day.

STEP 2/6

Step 1 — Check the Cache

Resolution starts locally and climbs only as needed.

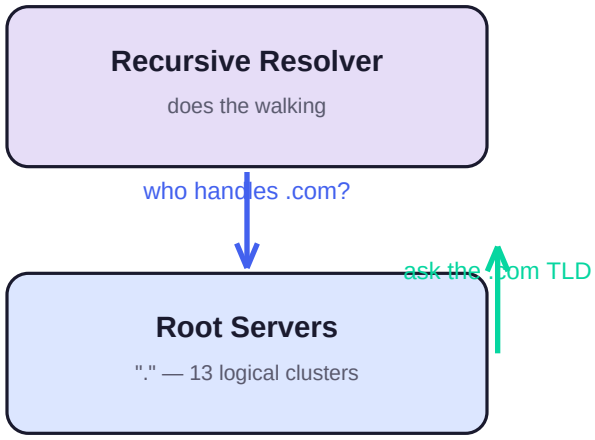


Most queries never leave the cache. TTLs control how long answers are kept — short TTL = flexible, long TTL = fast.

STEP 3/6

Step 2 — Ask the Root

On a cache miss, the recursive resolver walks the hierarchy top-down.



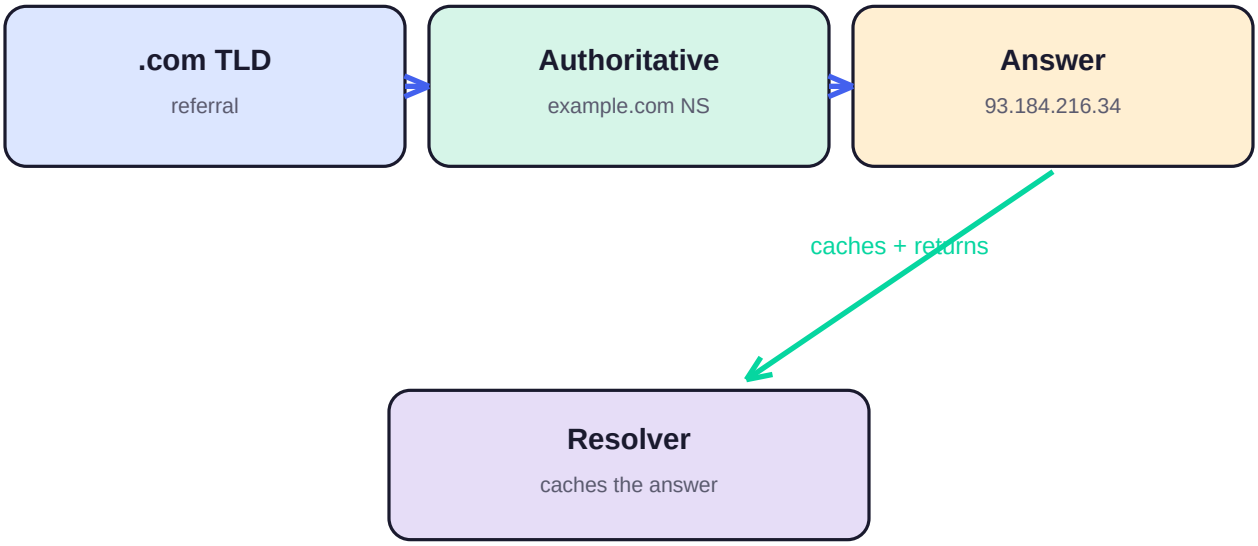
Root doesn't know example.com — but it knows who runs .com.

It answers with a REFERRAL, not the final answer.

STEP 4/6

Steps 3–4 — TLD, Then Authoritative

Each level refers down until someone is authoritative.



1. Resolver → .com TLD: 'who is authoritative for example.com?'
2. Resolver → authoritative NS: 'what is the A record?'
3. Authoritative answers; resolver caches it for the TTL.

STEP 5/6

Record Types: DNS Does More Than A Records

One system, many jobs.

A / AAAA

Hostname → IPv4 / IPv6

CNAME

Alias: www → example.com

MX

Where mail goes (+ priority)

TXT

SPF, DKIM, verification strings

NS / SOA

Delegation and zone authority

STEP 6/6

Security: DNS Is Traditionally Plaintext

Anyone on the path can see and spoof queries — hence the upgrades.

DNSSEC

Cryptographic signatures prove answers weren't tampered with (not encryption).

DoT (port 853)

DNS over TLS — encrypted channel to the resolver.

DoH (port 443)

DNS over HTTPS — queries blend in with web traffic.

Cache Poisoning

Attack: trick a resolver into caching a fake answer. DNSSEC defeats it.

STEP ✓/✓

Key Takeaways

DNS — Remember

- Resolution order: browser → OS → resolver cache → root → TLD → authoritative.
- Roots and TLDs give referrals; only authoritative servers give final answers.
- Everything is cached by TTL — DNS is fast because most queries never walk the tree.
- Record types: A/AAAA (addresses), CNAME (aliases), MX (mail), TXT (metadata), NS/SOA (delegation).
- Classic DNS is plaintext; DNSSEC adds authenticity, DoT/DoH add encryption.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev